PrepFusion

Electromagnetic Fields

Detailed Solutions • 2021–2026

GATE INSTRUMENTATION ENGINEERING

(IN)

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Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book works, in full, every question GATE has actually asked in Electromagnetic Fields (IN), gathered from the original papers and arranged unit by unit and year by year. Nothing is paraphrased or reconstructed: every question appears as it was set, and every solution is taken step by step rather than asserted.

Each solution carries the same identifier as the question it answers in the companion questions volume, so you can move between practice and explanation without a lookup table. Where the examining authority revised an answer or awarded marks to all (MTA), that is noted with the question it affects.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q3.24.7

3	Unit (chapter) number — Unit III of this book
24	GATE exam year — 2024 (two digits; 98 = 1998)
7	Question number within that unit and year

So **Q3.24.7** is the 7th question that GATE 2024 contributed to Unit III. Numbers are therefore not sequential through the book — they tell you the unit and the exam year at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a unit to see every question ID and click one to go straight to it.

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SOLUTIONS

I

Electrostatics

Topics

1. Coulomb's Law and Superposition Principle
2. Electric Field, Electric Field Intensity calculation in various Configurations: Due to Point, Line, Plane and Spherical Charge Distributions
3. Equipotential Surfaces and Electric Dipoles
4. Electric Flux Density and Gauss's Law
5. Divergence and Electric Potential
6. Electric Field Effect of Dielectric Medium- a)Boundary conditions b)method of image charges
7. Capacitance of Simple Configurations

Q1.26.1 MCQ 2026 1M Ans: MTA ● ● ○

The electric field vector points in the direction of the maximum decrease of the electric potential function. The electric field intensity is related to the potential by $\bar{E} = -\nabla V$:

$$\bar{E} = -\nabla V = -\left(\frac{\partial V}{\partial x}\hat{x} + \frac{\partial V}{\partial y}\hat{y} + \frac{\partial V}{\partial z}\hat{z}\right)$$

$$\bar{E} = -\left(\frac{x}{\sqrt{x^2+y^2+z^2}}\hat{x} + \frac{y}{\sqrt{x^2+y^2+z^2}}\hat{y} + \frac{z}{\sqrt{x^2+y^2+z^2}}\hat{z}\right)$$

At $(\hat{i}, \hat{j}, \hat{k})$ i.e., At $(1, 1, 1)$:

$$\bar{E} = -\frac{1}{\sqrt{3}}\hat{x} - \frac{1}{\sqrt{3}}\hat{y} - \frac{1}{\sqrt{3}}\hat{z}$$

$$\bar{F} = q\bar{E} = -\frac{1}{\sqrt{3}}\hat{x} - \frac{1}{\sqrt{3}}\hat{y} - \frac{1}{\sqrt{3}}\hat{z}$$

Comparing with $\bar{F} = a\hat{i} + b\hat{j} + c\hat{k}$:

$$a = -\frac{1}{\sqrt{3}}, \quad b = -\frac{1}{\sqrt{3}}, \quad c = -\frac{1}{\sqrt{3}}$$

$$\boxed{\text{MTA}}$$

Q1.26.2 MCQ 2026 2M Ans: A ● ○ ○

From Maxwell's first equation(Gauss's Law in differential form dictates that the divergence of the electric flux density field equals the volume charge density.):

$$\nabla \cdot \bar{D} = \rho_v \implies \rho_v = \varepsilon_o \nabla \cdot \bar{E}$$

Given the electric field vector components:

$$\nabla \cdot \bar{E} = \frac{\partial}{\partial x}(2x + 5y + 6z) + \frac{\partial}{\partial y}(5x + 4y + 10z) + \frac{\partial}{\partial z}(6x + 10y + 2z)$$

$$\nabla \cdot \bar{E} = 2 + 4 + 2 = 8$$

$$\rho_v = \varepsilon_o(8) = 8\varepsilon_o$$

$$\boxed{A)8\varepsilon_o}$$

Q1.25.1 MCQ 2025 1M Ans: A ● ○ ○

The electric flux density $\bar{D}$ due to an infinite sheet of charge is independent of the distance from the sheet and depends solely on the surface charge density.
Given surface charge density $\rho_s = 10\,\text{C/m}^2$.

$$\bar{E} = \frac{\rho_s}{2\varepsilon}\hat{r}$$

$$\bar{E} = \frac{\rho_s}{2\varepsilon}\hat{c}$$

$$\varepsilon\bar{E} = \frac{\rho_s}{2}\hat{c}$$

$$\bar{D} = \frac{10}{2}\hat{c}$$

$$\bar{D} = 5\hat{c}$$

$$\boxed{A)5\hat{c}}$$

Q1.25.2 NAT 2025 2M Ans: 4.00 to 4.40 ● ● ○

Electrostatic force between the plates is given by:

$$F_e = \frac{Q^2}{2\varepsilon A}$$

At Equilibrium, the mechanical spring restoring force balances the electrostatic attraction force.The restoring spring force is $F_s = kx$. At equilibrium position:

$$F_s = F_e \implies kx = \frac{Q^2}{2\varepsilon A}$$

$$Q^2 = 2\varepsilon Akx \implies Q = \sqrt{2\varepsilon Akx}$$

Substituting the given parameters:

$$Q = \sqrt{2 \times 8.854 \times 10^{-12} \times (100 \times 10^{-6})^2 \times 0.01 \times 10^{-6}}$$

$$Q = \sqrt{2 \times 8.854 \times 10^{-30}}$$

$$Q = \sqrt{1.7708 \times 10^{-29}} = 4.208 \times 10^{-14}\,\text{C}$$

$$\boxed{4.20}$$

Q1.24.1 MCQ 2024 1M Ans: A ● ○ ○

The capacitance of a spherical shell arrangement depends on the geometric radii of the concentric spheres and the permittivity of the dielectric medium. The capacitance of a spherical capacitor with inner radius a and outer radius b is given by:

$$C = \frac{4\pi\varepsilon}{\left[\frac{1}{a} - \frac{1}{b}\right]} = \frac{4\pi\varepsilon ab}{(b-a)}$$

Given $a = x$ and $b = y$:

$$C = \left[\frac{4\pi\varepsilon xy}{y-x}\right]$$

$$\boxed{A)\frac{4\pi\varepsilon xy}{y-x}}$$

Q1.24.2 MCQ 2024 2M Ans: B ● ● ○

Given total current $I = \iint \bar{J} \cdot d\bar{A} = \iint ks^2\,\rho d\rho d\phi$ where $(s = \rho)$ represents the radial distance:

$$I = \iint k \cdot \rho^2 \cdot \rho d\rho d\phi = k \int \rho^3 d\rho \int d\phi$$

$$= k\left[\frac{\rho^4}{4}\right]_0^a [\phi]_0^{2\pi}$$

$$= \frac{k}{4}\left[a^4 - 0\right][2\pi]$$

$$I = \frac{\pi k a^4}{2}$$

$$\boxed{B)\frac{\pi k a^4}{2}}$$

Q1.22.1 NAT 2022 2M Ans: 33.3 ● ● ○

Given: $\varepsilon_o = 8.854 \times 10^{-12}$, $a = 20\,\text{cm}$, $b = 25\,\text{cm}$.
The capacitance C of a spherical capacitor is:

$$C = \frac{4\pi\varepsilon_o}{\left[\frac{1}{a} - \frac{1}{b}\right]} = \frac{4\pi\varepsilon_o ab}{b-a}$$

$$C = \frac{4\pi \times \frac{1}{36\pi \times 10^9} \times 15 \times 10 \times 10^{-4}}{5 \times 10^{-2}}$$

$$C = \frac{1}{9 \times 10^9} \times 30 \times 10^{-2}$$

$$C = \frac{1}{3} \times 10^{-1} \times 10^{-9} = \frac{1}{3} \times 10^{-10}$$

$$C = 33.3\,\text{pF}$$

$$\boxed{33.3}$$

Key Points

- The capacitance of concentric spherical conductors depends only on permittivity and geometric boundaries.
- Free space permittivity can be approximated as $\varepsilon_o \approx \frac{1}{36\pi \times 10^9}$ F/m.

Q1.21.1 MCQ 2021 1M Ans: B ● ○ ○

If an infinite line lies along the Z-axis, carrying a uniform charge density ρ_l C/m, then the electric field intensity in the cylindrical coordinate system (r, ϕ, z) is given by:

$$\bar{E} = \frac{\rho_l}{2\pi\varepsilon r}\hat{a}_r \quad \text{or} \quad \frac{\rho_l}{2\pi\varepsilon r}\hat{r}\ \text{V/m}$$

Therefore, the electric field intensity will be along the $\hat{r}$ direction only. The direction of the field is always radially outwards from the line charge.

$$\boxed{\text{B) } \hat{r}}$$

DEBUG INFO

Chapter I

Electrostatics

Total Questions : 8

MCQ : 6

MSQ : 0

NAT : 2

SOLUTIONS

II

Magnetostatics

Topics

1. Biot-Savart's Law
2. Electromagnetic Induction and Faraday's Law
3. Magnetic Field of a current carrying loop
4. Ampere's Law and Curl
5. Lorentz Force
6. Inductance and Magnetomotive Force
7. Self and Mutual Inductance of Simple Configurations
8. Reluctance and Magnetic Circuits
9. Magnetic Flux and Boundary Conditions for magnetic field

Q2.26.1 MCQ 2026 2M Ans: C ● ● ○

The differential force equation yields:

$$\bar{F} = Id\bar{\ell}\left(\frac{1}{\sqrt{6}}\hat{x} + \frac{2}{\sqrt{6}}\hat{y} + \frac{1}{\sqrt{6}}\hat{z}\right) \times (2\hat{x} + 2\hat{z})$$

Evaluating the cross product using determinants:

$$= \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 2 & 0 & 2 \end{vmatrix}$$

$$= \hat{x}\left(\frac{4}{\sqrt{6}} - 0\right) - \hat{y}\left(\frac{2}{\sqrt{6}} - \frac{2}{\sqrt{6}}\right) + \hat{z}\left(0 - \frac{4}{\sqrt{6}}\right)$$

$$= \frac{4}{\sqrt{6}}\hat{x} - \frac{4}{\sqrt{6}}\hat{z}$$

$$\bar{F} = d\bar{\ell}\left(\frac{4}{\sqrt{6}}\hat{x} - \frac{4}{\sqrt{6}}\hat{z}\right)$$

$$\frac{\bar{F}}{d\bar{\ell}} = \left(\frac{4}{\sqrt{6}}\hat{x} - \frac{4}{\sqrt{6}}\hat{z}\right)$$

$$\boxed{\text{C) } \frac{4}{\sqrt{6}}\hat{x} - \frac{4}{\sqrt{6}}\hat{z}}$$

Key Points

- The magnetic force on a differential current element is computed using the vector cross product of the direction vector and the magnetic field vector.

Q2.25.1 MCQ 2025 1M Ans: A ● ● ○

The magnetic force on a current-carrying conductor is maximized when the conductor is perpendicular to the magnetic field lines.
Given:

$$I = \frac{60}{0.5} = 120\,\text{A}$$

$$F = BIl\sin\theta \quad (\theta = 90^\circ)$$

$$F = BIl = 0.5 \times 1 \times 120 = 60\,\text{N}$$

Using the Right-Hand Rule, the direction is in the positive x-direction.

$$\boxed{\text{A) } 60N \text{ in the direction of } \hat{x}}$$

Q2.23.1 NAT 2023 1M Ans: 5 ● ○ ○

Magnetic force per unit length between two parallel infinitely long current-carrying conductors, spaced 'd' apart is given by:

$$f = \frac{\mu I_1 I_2}{2\pi d}$$

$$f \propto \frac{1}{d}$$

$$\frac{f_2}{f_1} = \frac{d_1}{d_2} \implies f_2 = \left(\frac{d_1}{d_2}\right) f_1$$

Given: $d_1 = 2\,\text{cm}$ and $f_1 = 10\,\mu\text{N/m}$

$$d_2 = 2(d_1) = 2 \times 2\,\text{cm}$$

$$f_2 = \left(\frac{2}{2 \times 2}\right)(10)$$

$$\therefore\ f_2 = 5\,\mu\text{N/m}$$

$$\boxed{5}$$

Q2.22.1 NAT 2022 2M Ans: 200 ● ● ●

Given: Radius $= 0.1$ m, $\omega = 20$ rad/sec, $B = 2$ T

$$T = \frac{2\pi}{\omega}, \quad V = -\frac{d\psi}{dt}$$

$$\frac{T}{2} = \frac{\pi}{\omega}$$

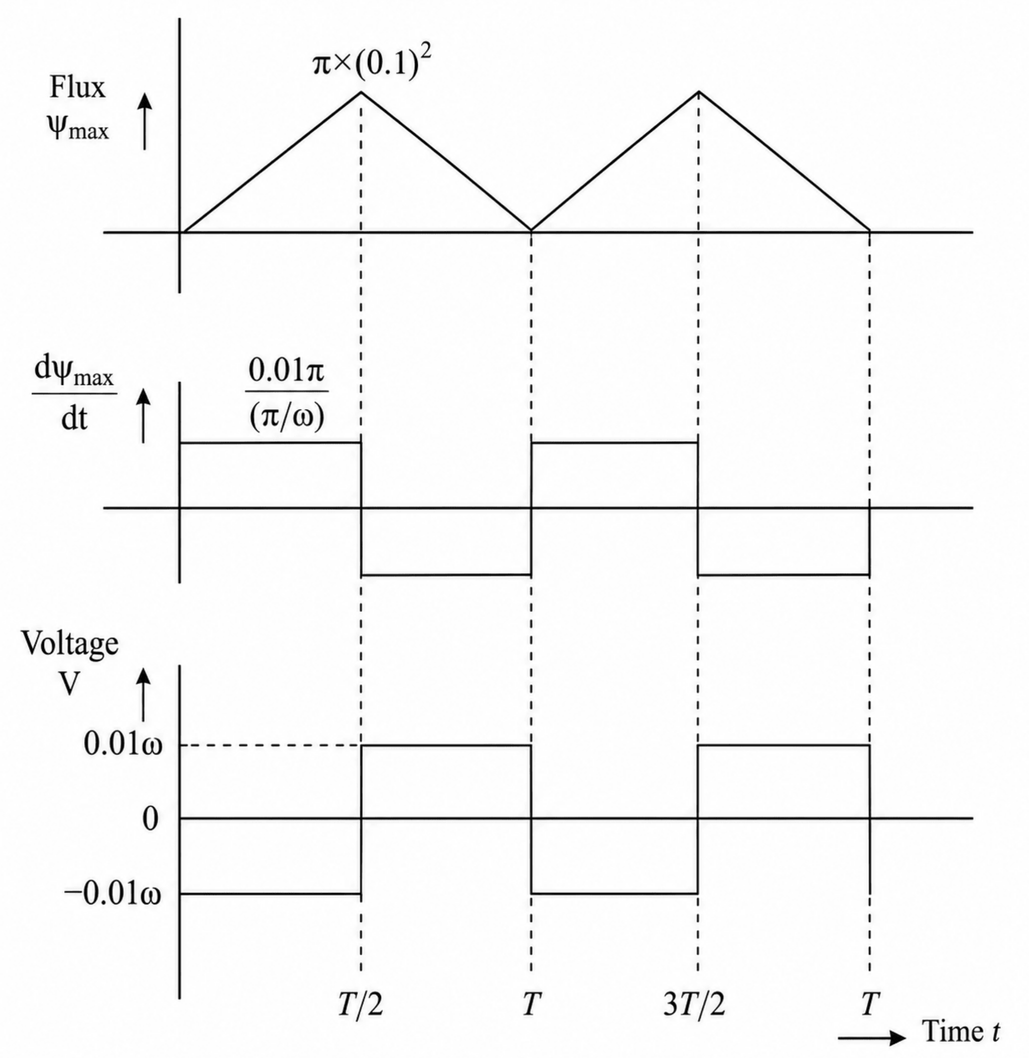

Fig. Solution Q2.22.1

$$V_{p-p} = 0.02\omega = 0.02 \times 20 = 0.4$$

$$I_{p-p} = \frac{0.4}{2} = 0.2 = 200 \times 10^{-3}$$

$$I_{p-p} = 200\,\text{mA}$$

$$\boxed{200}$$

Key Points

- Induced voltage is proportional to the time rate of change of magnetic flux linkage.
- Peak-to-peak values can be evaluated directly from the extrema transitions of the derivative waveforms.

Q2.21.1 | NAT | 2021 | 2M | Ans: 2.7 to 2.9 | ● ● ●

The net force (F) on the bar, which is subject to friction, is equal to the sum of the applied force (Electromagnetic Lorentz Force or F_{appl}) and the frictional force (F_{fr}).

$$F = F_{\text{appl}} - F_{\text{fr}}$$

(The negative sign is due to the frictional force opposing motion)

$$ma = IlB - kmg$$

Where: m : mass of bar 1 kg, I : current, $I = 10\,\text{A}$, l : length of bar $l = 0.5\,\text{m}$, $B = 1$ T, k : Coefficient of friction $k = 0.1$, $g = 10\,\text{m/sec}^2$,

$$a = \frac{IlB}{m} - kg = \frac{10 \times 0.5 \times 1}{1} - 0.1 \times 10$$

$$a = 4\,\text{m/sec}^2$$

Acceleration, $a = 4\,\text{m/sec}^2$ From basics of Kinematics:

$$v^2 - u^2 = 2as$$

Where 's' is the distance travelled by the bar, $s = 1\,\text{m}$. As the bar is initially resting on the surface, its initial velocity $u = 0$.

$$v^2 = 2 \times (4) \times (1)$$

$$\therefore \text{ Speed, } v = 2.82\,\text{m/sec}$$

$$\boxed{2.82}$$

Q2.21.2 | NAT | 2021 | 2M | Ans: 10 | ● ● ○

Given: Toroid is made of CRGO with $\mu_r = 3000$, $\mu_0 = 4\pi \times 10^{-7}$ H/m, $I = 5\,\text{mA}$, $N = 200$, Inner diameter, $D_1 = 10\,\text{cm}$ Outer diameter, $D_2 = 14\,\text{cm}$ Mean diameter:

$$D = \left(\frac{10 + 14}{2}\right) = 12\,\text{cm}$$

Mean radius, $R = 6\,\text{cm}$ Magnetic flux density in the core is given by:

$$B = \frac{\mu N I}{l}$$

Where l is the mean length of the toroid, $l = 2\pi R$.

$$B = \frac{\mu_0 \mu_r N I}{2\pi R}$$

$$B = \frac{4\pi \times 10^{-7} \times 3000 \times 200 \times 5 \times 10^{-3}}{2\pi \times 6 \times 10^{-2}}$$

$$B = \frac{2 \times 10^{-7} \times 3 \times 10^{3} \times 2 \times 10^{2} \times 5 \times 10^{-3}}{6 \times 10^{-2}}$$

$$B = 10\,\text{mT}$$

$$\boxed{10}$$

DEBUG INFO

Chapter II

Magnetostatics

Total Questions : 6

MCQ : 2

MSQ : 0

NAT : 4